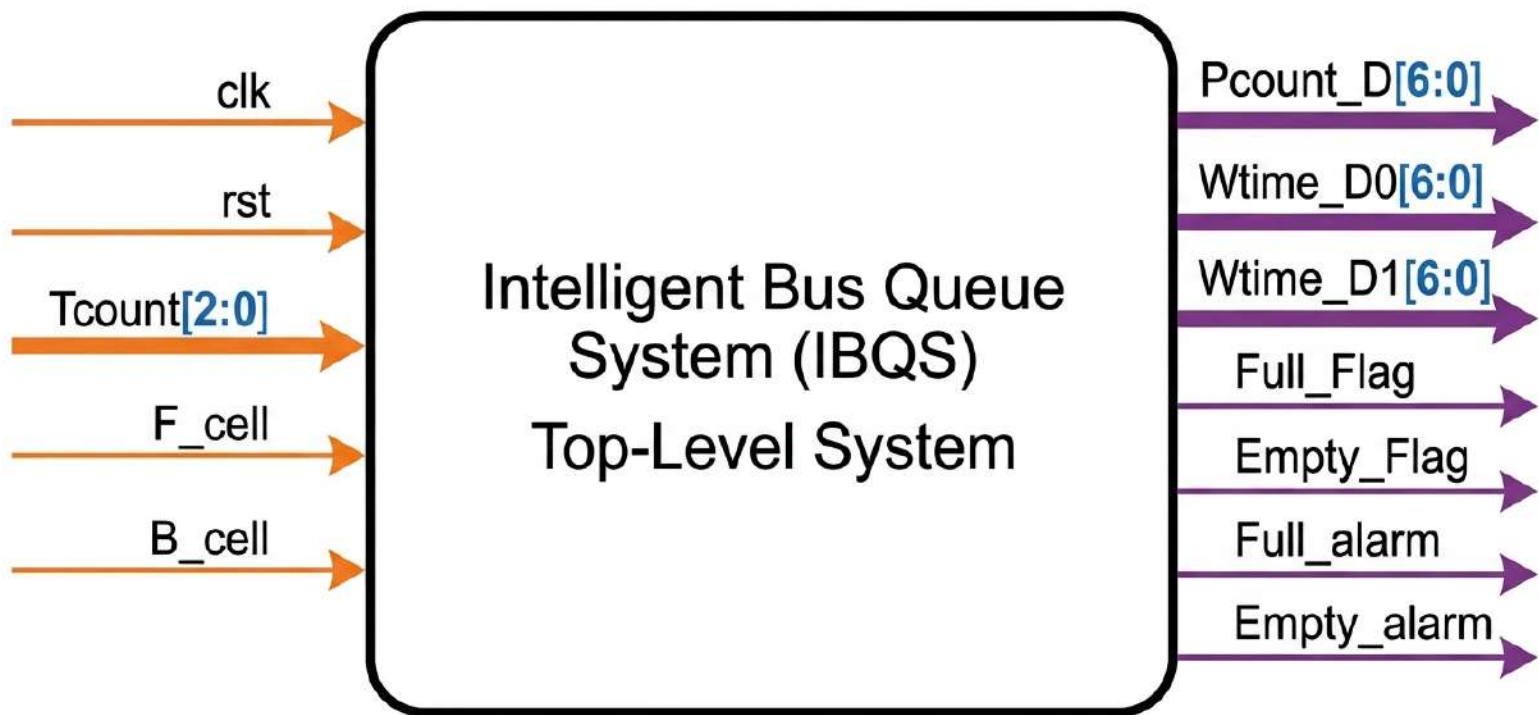
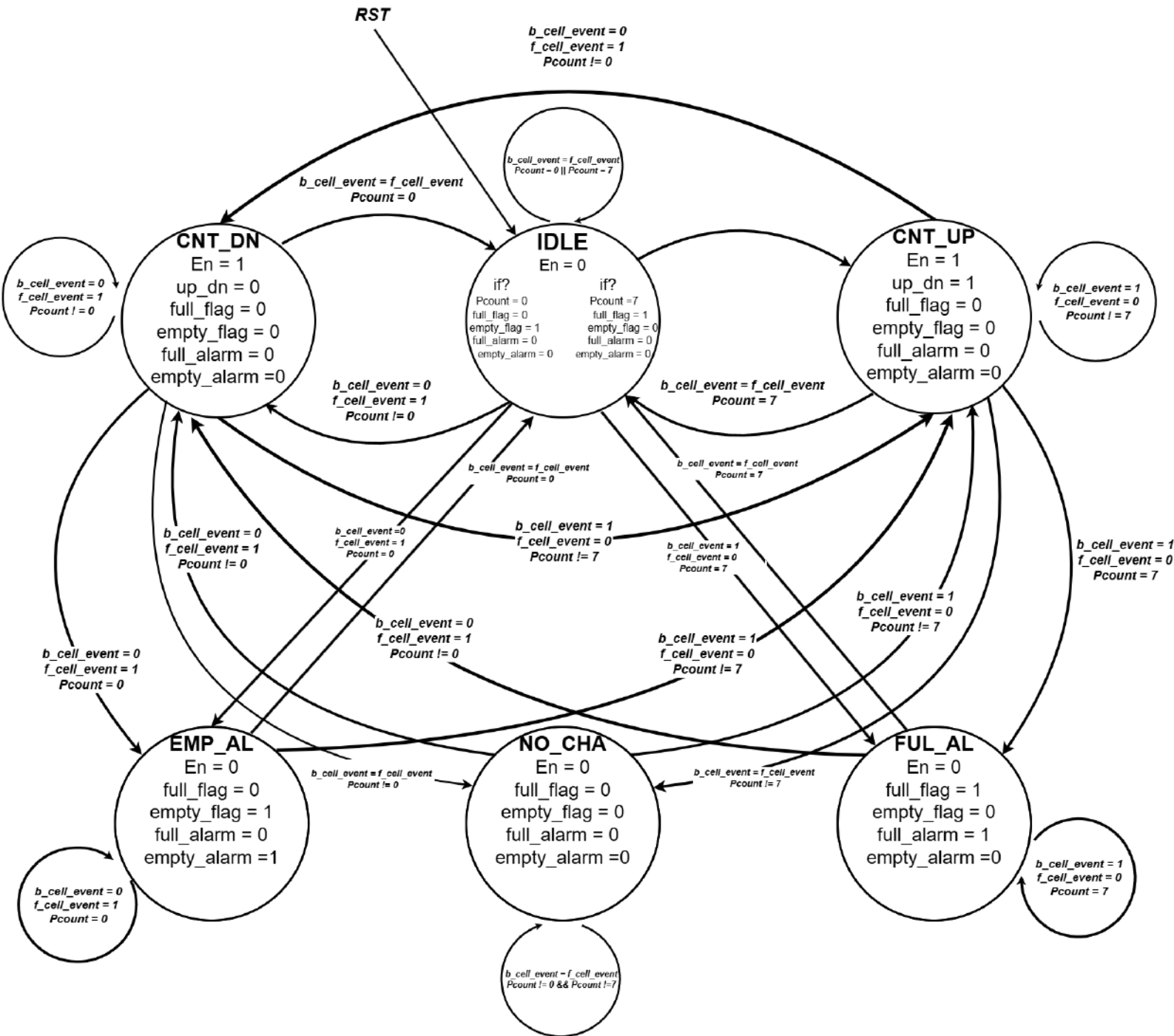


I/O INTERFACE SPECIFICATION

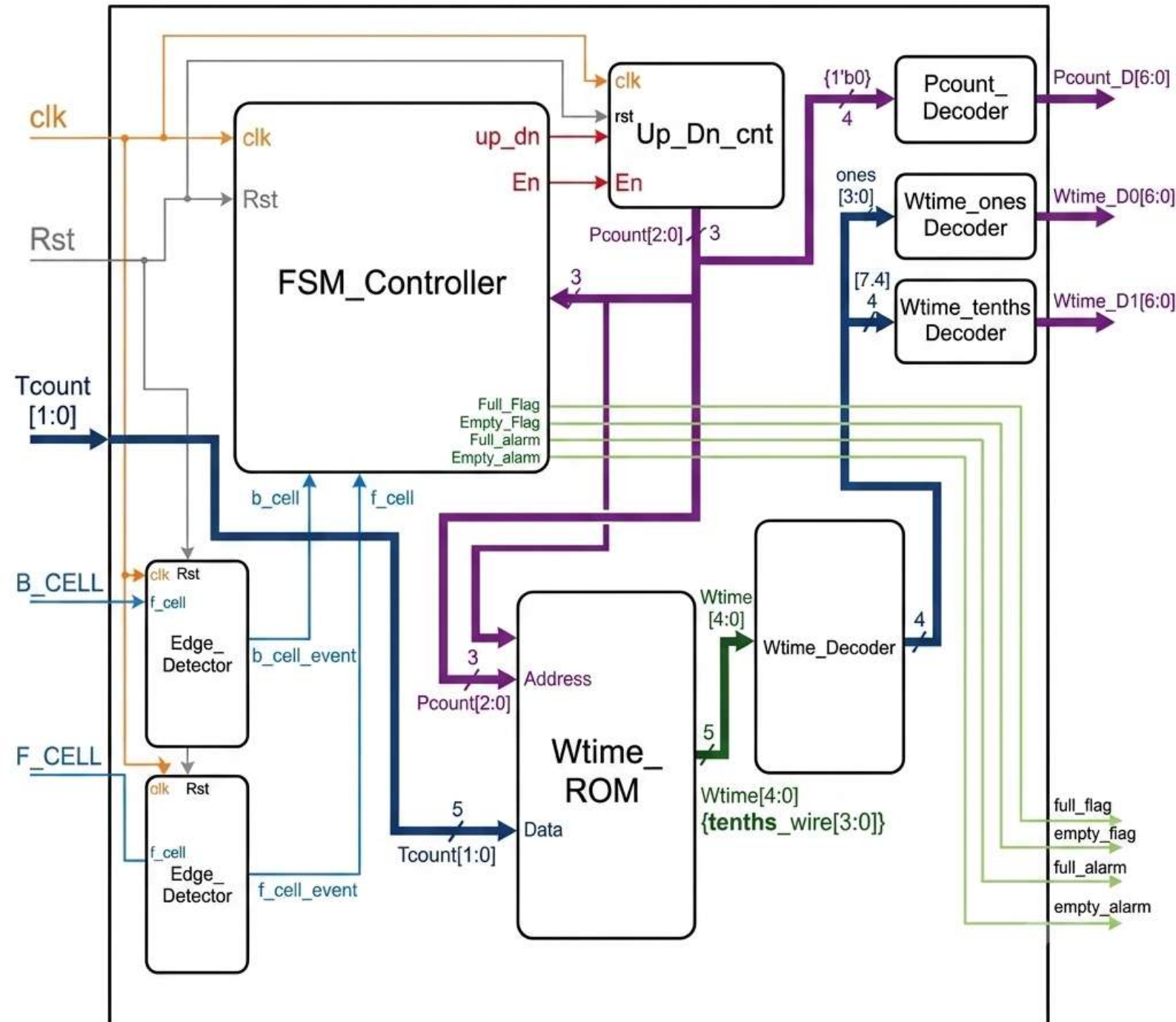
Signal Name	Direction	Width	Data Type	Description	Valid Values
clk	Input	1-bit	Clock	System clock used to synchronize all internal sequential registers (Edge Detectors, FSM Controller, and Up/Down Counter).	0 or 1 (Continuous oscillation)
rst	Input	1-bit	Reset	Active-low system reset. Asynchronously initializes the state registers, edge detectors, and counter to their default IDLE/zero states.	0 (Reset active), 1 (Normal operation)
Tcount	Input	2-bit [1:0]	Control Bus	Configures the number of active service tellers. Used by the internal lookup table (Wtime_ROM) to dynamically scale wait times.	2'b01 (1 Teller) 2'b10 (2 Tellers) 2'b11 (3 Tellers)
F_CELL	Input	1-bit	Sensor Input	Front photoelectric sensor detecting customer entry/exit. An active-low edge-triggered input (transition from high to low) signals a physical event.	0 (Sensor blocked / Event) 1 (Sensor clear)
B_CELL	Input	1-bit	Sensor Input	Rear photoelectric sensor detecting customer entry/exit. An active-low edge-triggered input (transition from high to low) signals a physical event.	0 (Sensor blocked / Event) 1 (Sensor clear)
Pcount_D	Output	7-bit [6:0]	Seven-Segment	Active-high segment output representing the current customer occupancy count in the queue. Displays digits from 0 to 7.	7'b111_1110 (Digit 0) to 7'b111_0000 (Digit 7)
Wtime_D0	Output	7-bit [6:0]	Seven-Segment	Active-high segment output displaying the units (ones) digit of the calculated waiting time (derived from decoded BCD values).	7'b111_1110 (Digit 0) to 7'b111_1011 (Digit 9)
Wtime_D1	Output	7-bit [6:0]	Seven-Segment	Active-high segment output displaying the tens digit of the calculated waiting time (derived from decoded BCD values).	7'b111_1110 (Digit 0) to 7'b110_1101 (Digit 2)
FULL_FLAG	Output	1-bit	Status Flag	Active-high status flag asserted when the queue reaches its maximum capacity of 7 people (Pcount = 3'd7).	0 (Capacity available) 1 (Queue full)
EMPTY_FLAG	Output	1-bit	Status Flag	Active-high status flag asserted when the queue has no customers (Pcount = 3'd0).	0 (Queue has customers) 1 (Queue empty)
FULL_ALARM	Output	1-bit	Alarm	Active-high alarm signal activated when a customer entry event occurs while the queue is already full.	0 (Normal operation) 1 (Alarm active)
EMPTY_ALARM	Output	1-bit	Alarm	Active-high alarm signal activated when a customer exit event occurs while the queue is already empty.	0 (Normal operation) 1 (Alarm active)

System Block Diagram





Structural Block Diagram



ITI BQS – TEST PLAN (Implemented in Testbench)

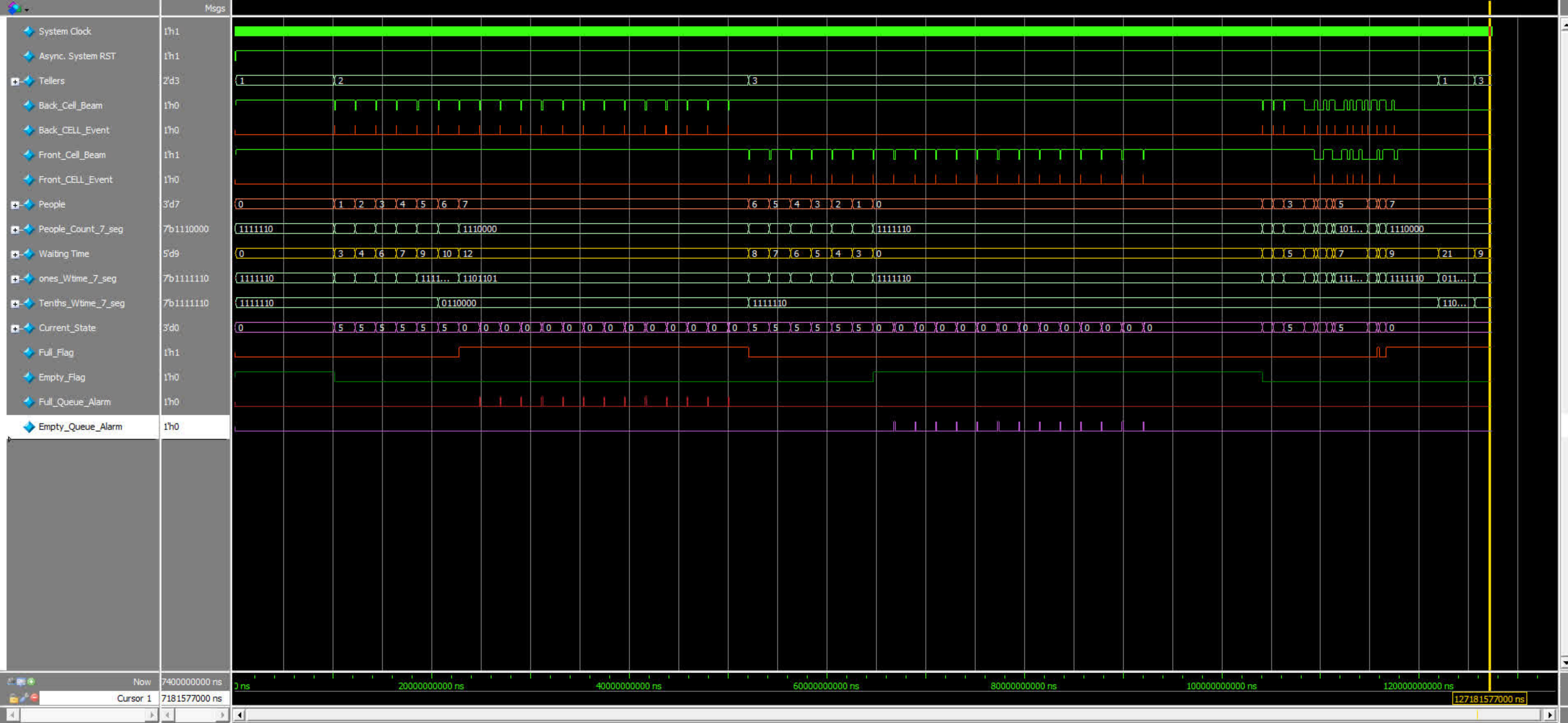
Test Case	Scenario	Tcount (Tellers)	Stimulus (Events)	Expected Behavior	Check / Observations
1. RESET TEST					
TC_01	System Reset	2 (2'b10)	- Assert rst = 0 for few cycles - Deassert rst = 1	- Pcount = 0 - All flags = 0 - All alarms = 0 - Displays show 0	Verify initial values after reset
2. COUNTING UP (CUSTOMER ENTRY)					
TC_02	Single Entry	2 (2'b10)	Trigger F_cell event (entry)	- Pcount increments by 1 (0 → 1) - No flags/alarm active	Pcount_D = 1
TC_03	Multiple Entries	2 (2'b10)	Trigger F_cell event 5 times	- Pcount increments each time (0 → 5) - No flags/alarm active	Pcount_D = 5
TC_04	Fill to Capacity	2 (2'b10)	Trigger F_cell until Pcount = 7	- Pcount = 7 - FULL_FLAG = 1	Pcount_D = 7 FULL_FLAG = 1
3. COUNTING DOWN (CUSTOMER EXIT)					
TC_05	Single Exit	2 (2'b10)	Trigger B_cell event (exit)	- Pcount decrements by 1 (7 → 6) - FULL_FLAG = 0	Pcount_D = 6 FULL_FLAG = 0
TC_06	Multiple Exits	2 (2'b10)	Trigger B_cell event 4 times	- Pcount decrements each time (6 → 2) - No flags/alarm active	Pcount_D = 2
TC_07	Empty the Queue	2 (2'b10)	Trigger B_cell until Pcount = 0	- Pcount = 0 - EMPTY_FLAG = 1	Pcount_D = 0 EMPTY_FLAG = 1
4. ALARM CONDITIONS					
TC_08	Entry When Full	2 (2'b10)	- Ensure Pcount = 7 - Trigger F_cell event	- Pcount remains = 7 - FULL_ALARM = 1	FULL_ALARM = 1
TC_09	Exit When Empty	2 (2'b10)	- Trigger Pcount = 0 - Trigger B_cell event	- Pcount remains = 0 - EMPTY_ALARM = 1	EMPTY_ALARM = 1
5. DIFFERENT TELLER CONFIGURATIONS (WAITING TIME CHECK)					
TC_10	1 Teller	1 (2'b01)	- Set Tcount = 1 - Set Pcount = 5	- Waiting time = ROM[1][5] - Wtime_D1, Wtime_D0 show correct value	Verify on 7-seg displays
TC_11	2 Tellers	2 (2'b10)	- Set Tcount = 2 - Set Pcount = 5	- Waiting time = ROM[2][5] - Wtime_D1, Wtime_D0 show correct value	Verify on 7-seg displays
TC_12	3 Tellers	3 (2'b11)	- Set Tcount = 3 - Set Pcount = 5	- Waiting time = ROM[3][5] - Wtime_D1, Wtime_D0 show correct value	Verify on 7-seg displays
6. STRESS TEST					
TC_13	Random Operations	2 (2'b10)	Randomly apply F_cell and B_cell events for many cycles	- Pcount always within 0..7 - Flags and alarms behave correctly	No overflow/underflow Stable operation

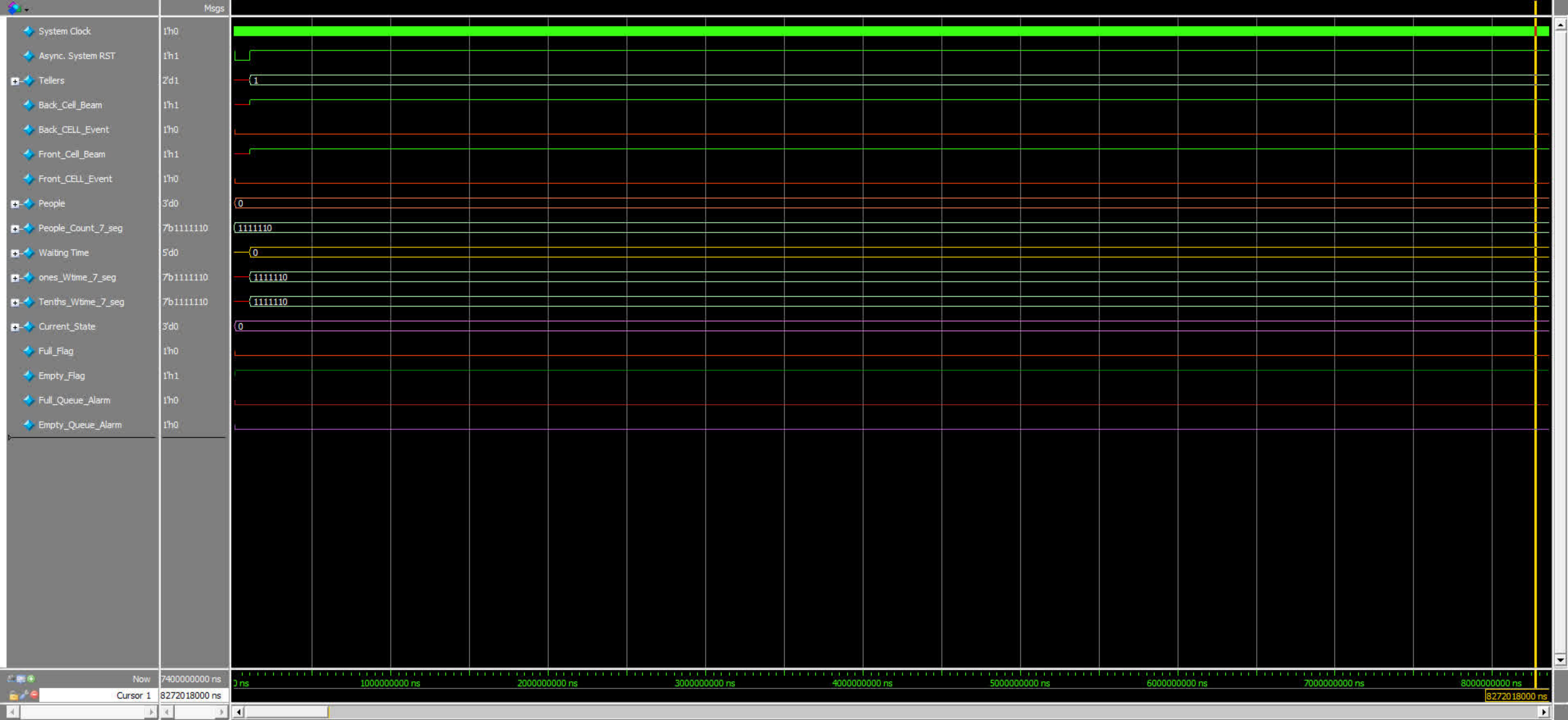
NOTES:

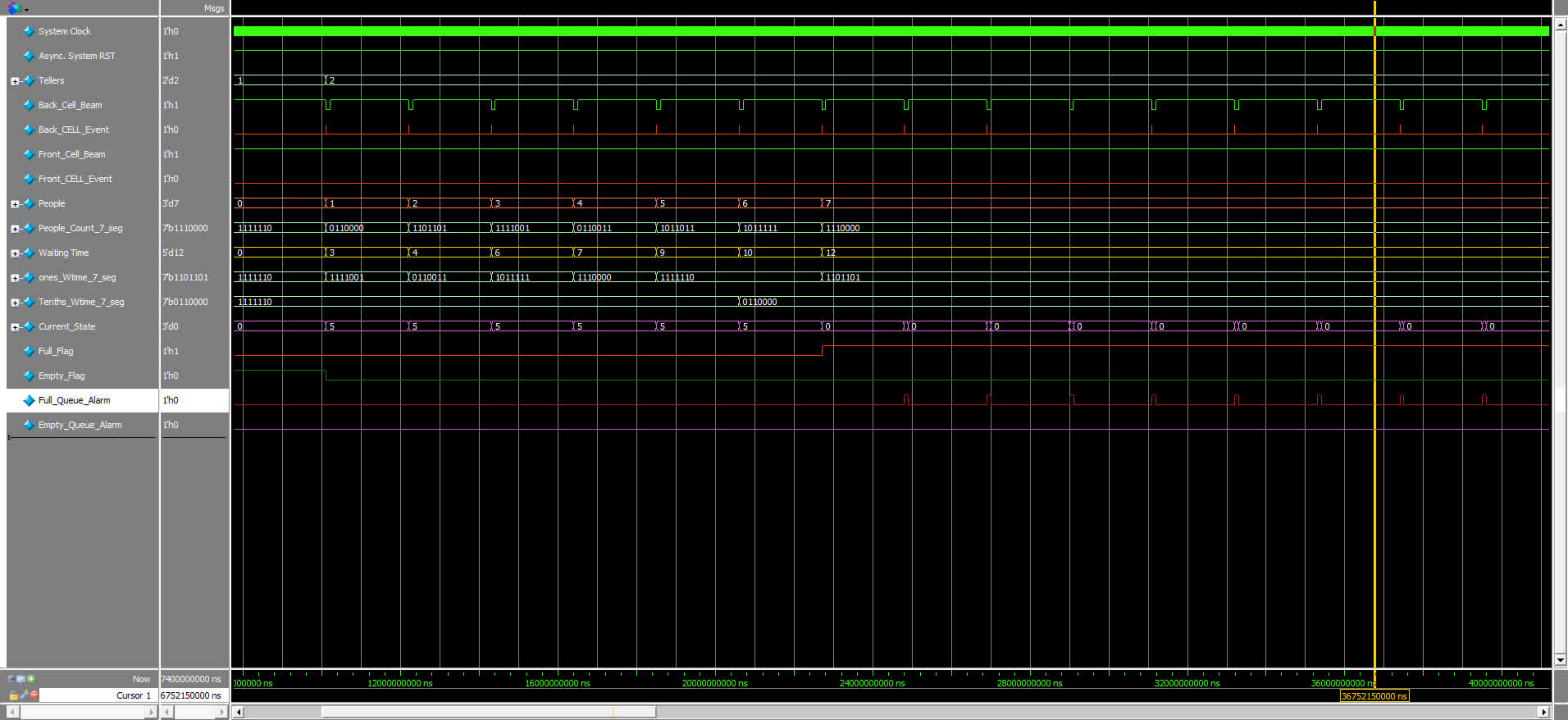
- Events (F_cell, B_cell) are edge-detected (active-low).
- Small delays are inserted between events in the testbench to ensure proper edge detection.
- All outputs are monitored and compared with expected results.

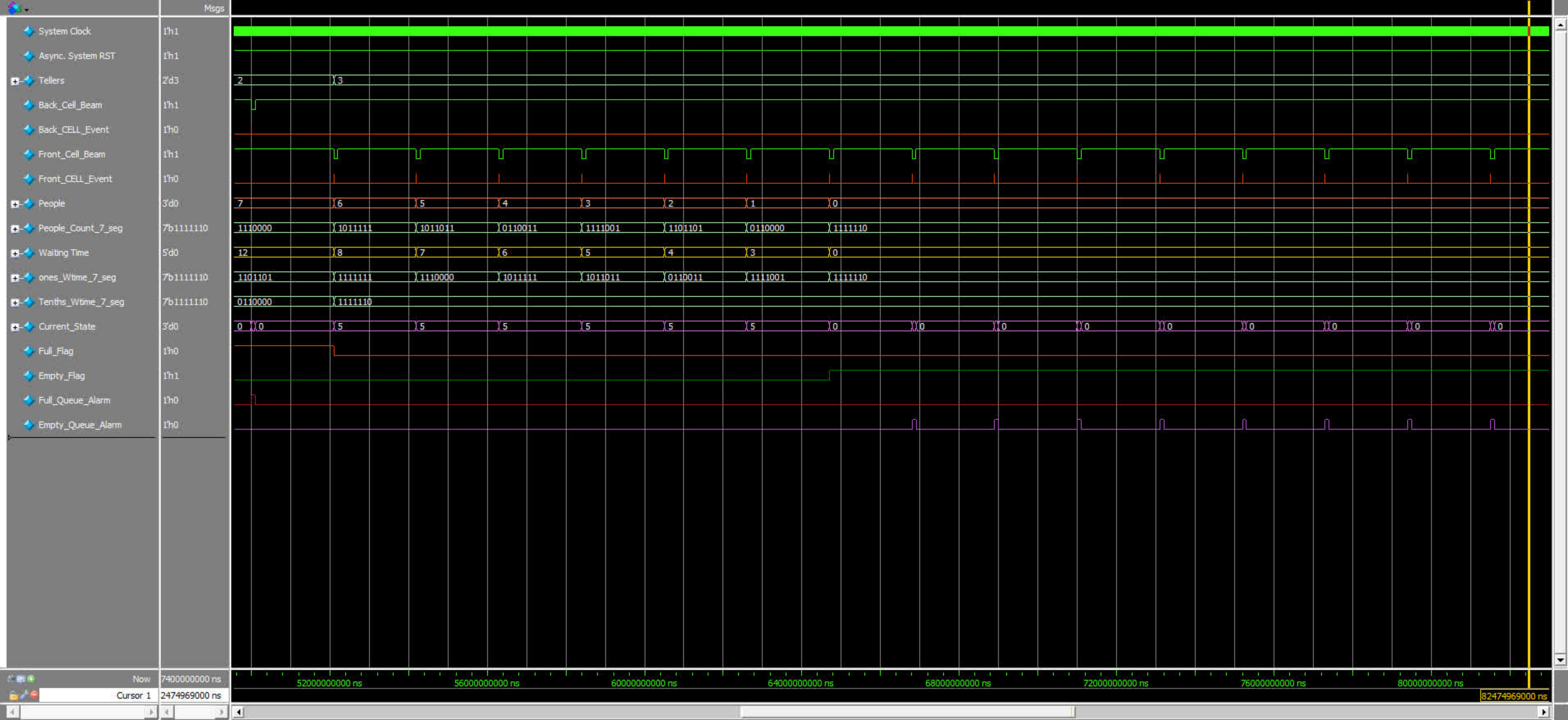
Signal Legend:

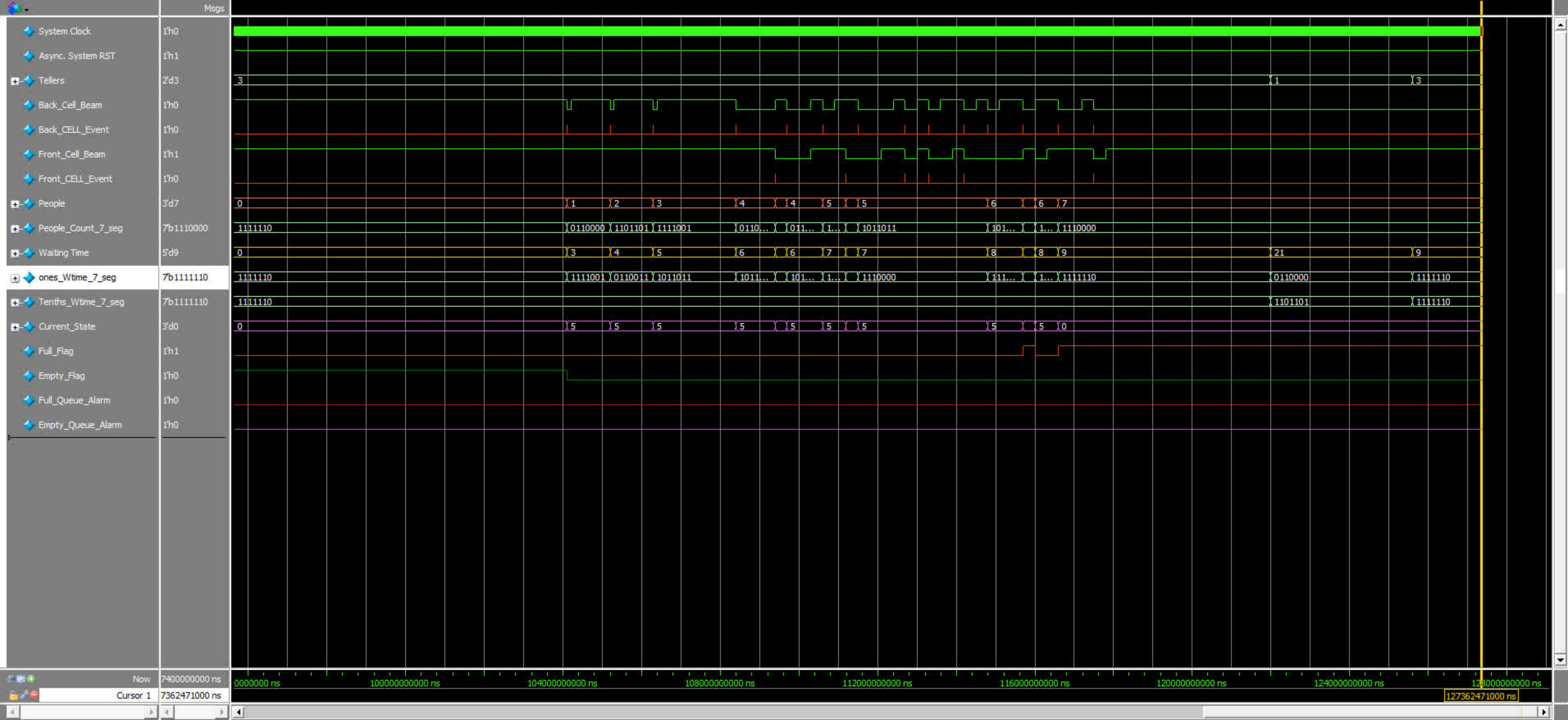
F_cell	= Front photoelectric sensor (Entry)	FULL_FLAG	= Queue Full indicator
B_cell	= Back photoelectric sensor (Exit)	EMPTY_FLAG	= Queue Empty indicator
Pcount	= People count (0 to 7)	FULL_ALARM	= Alarm when entry at full
		EMPTY_ALARM	= Alarm when exit at empty

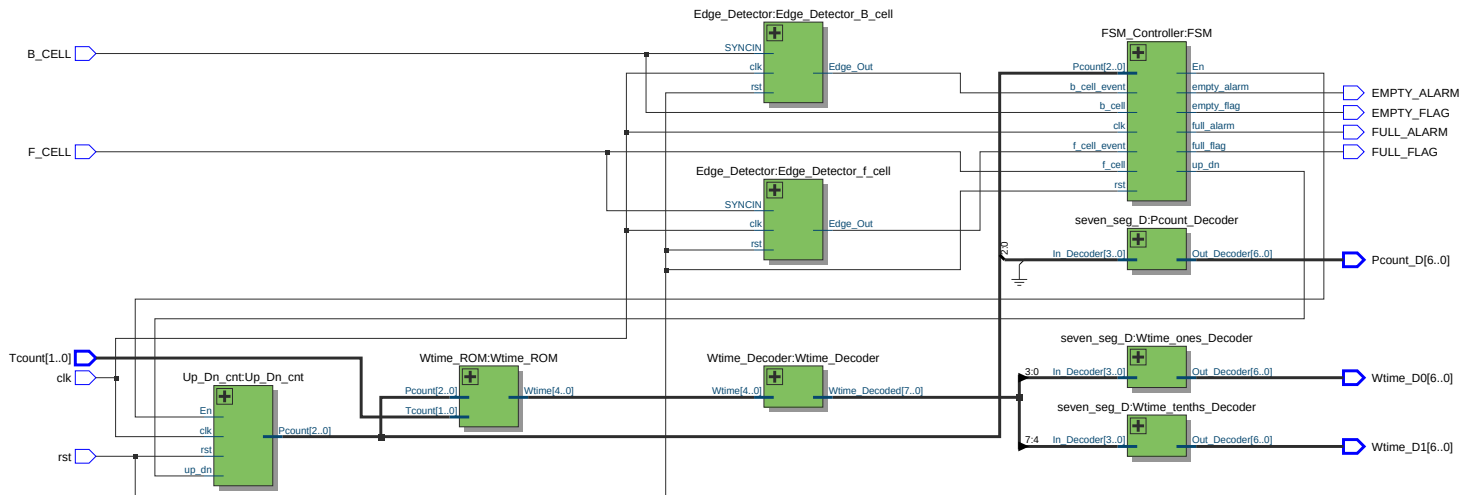


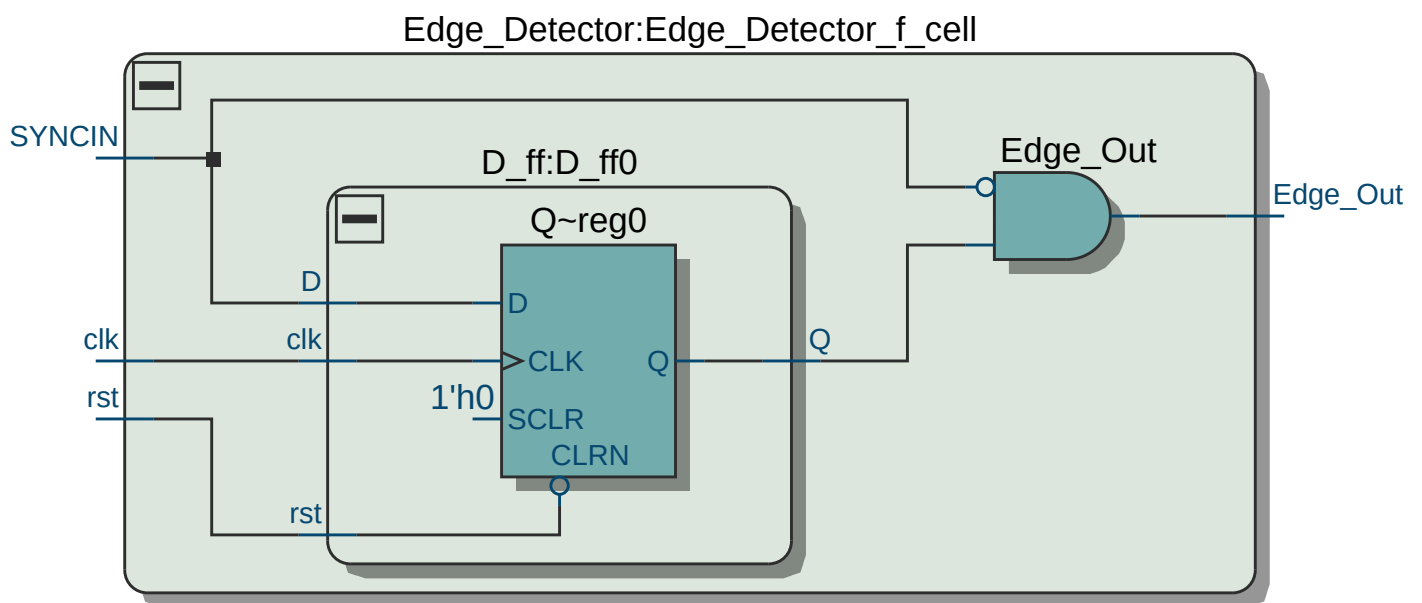




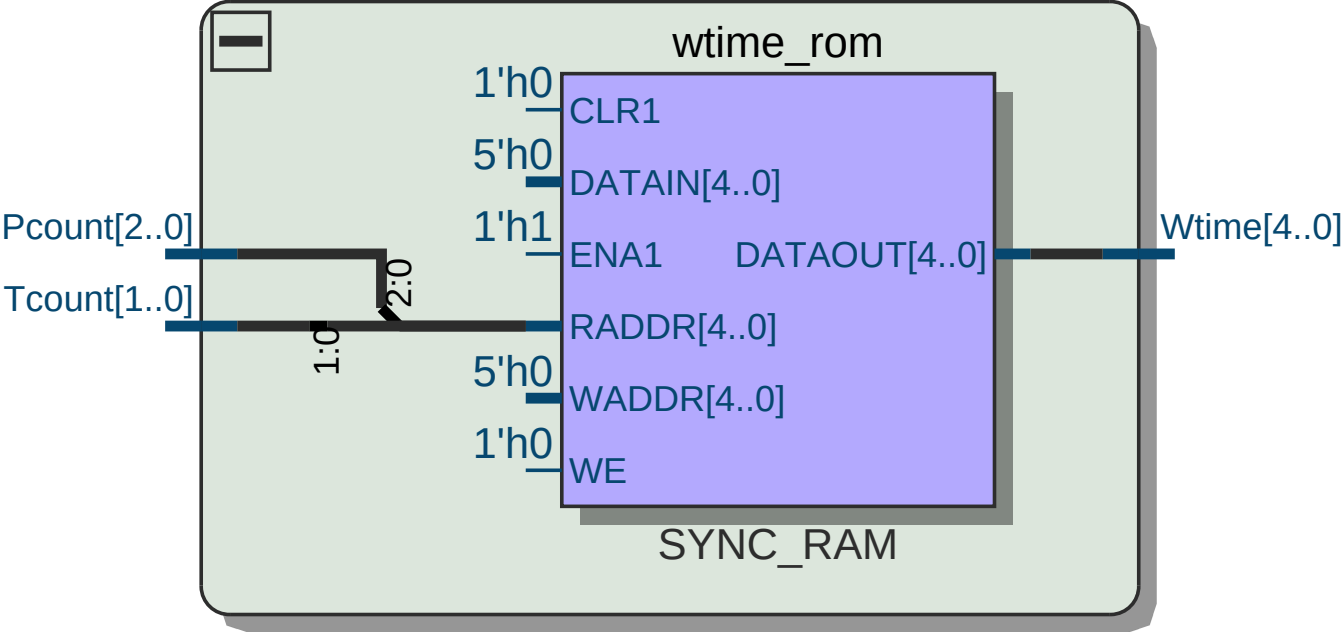




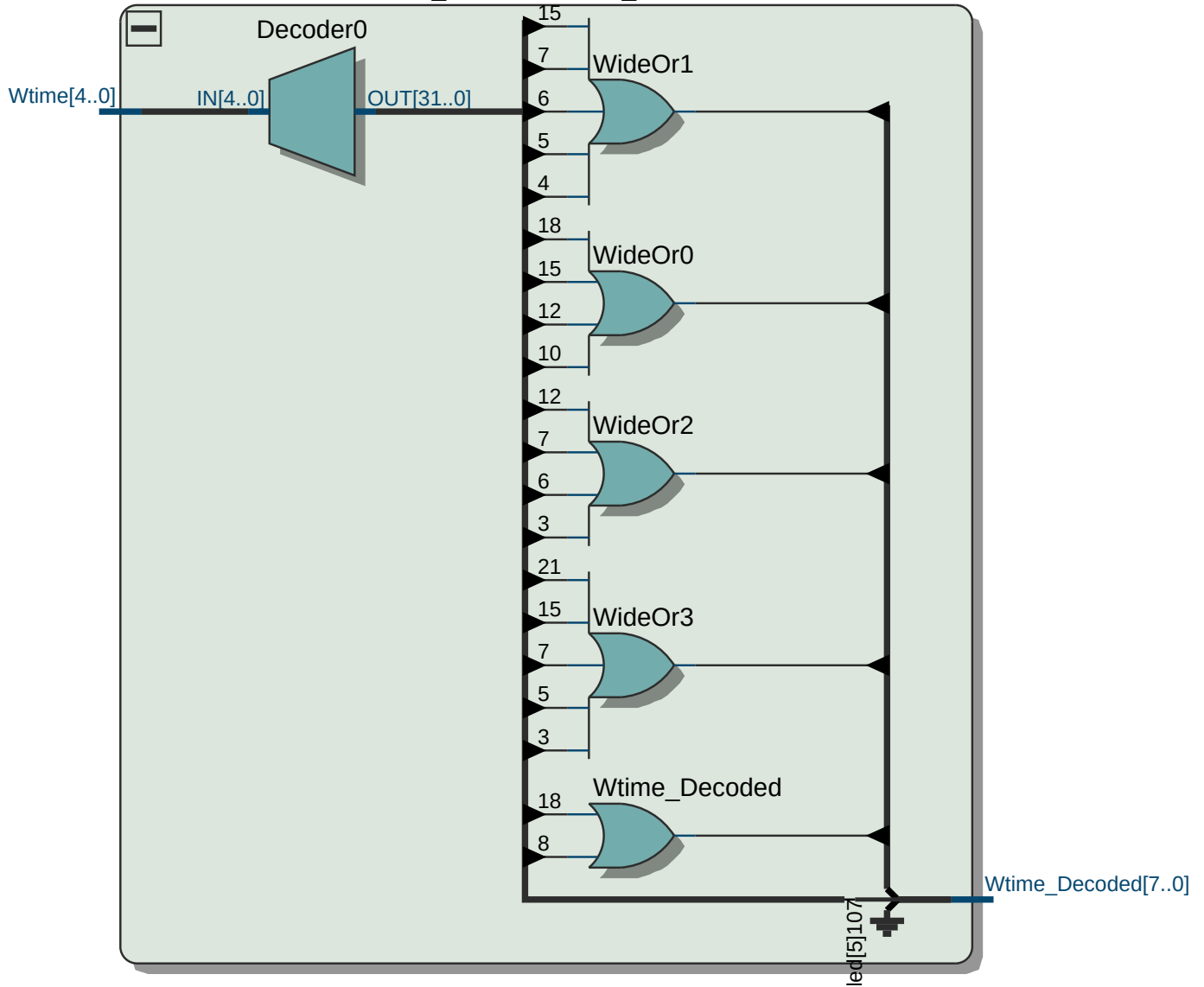




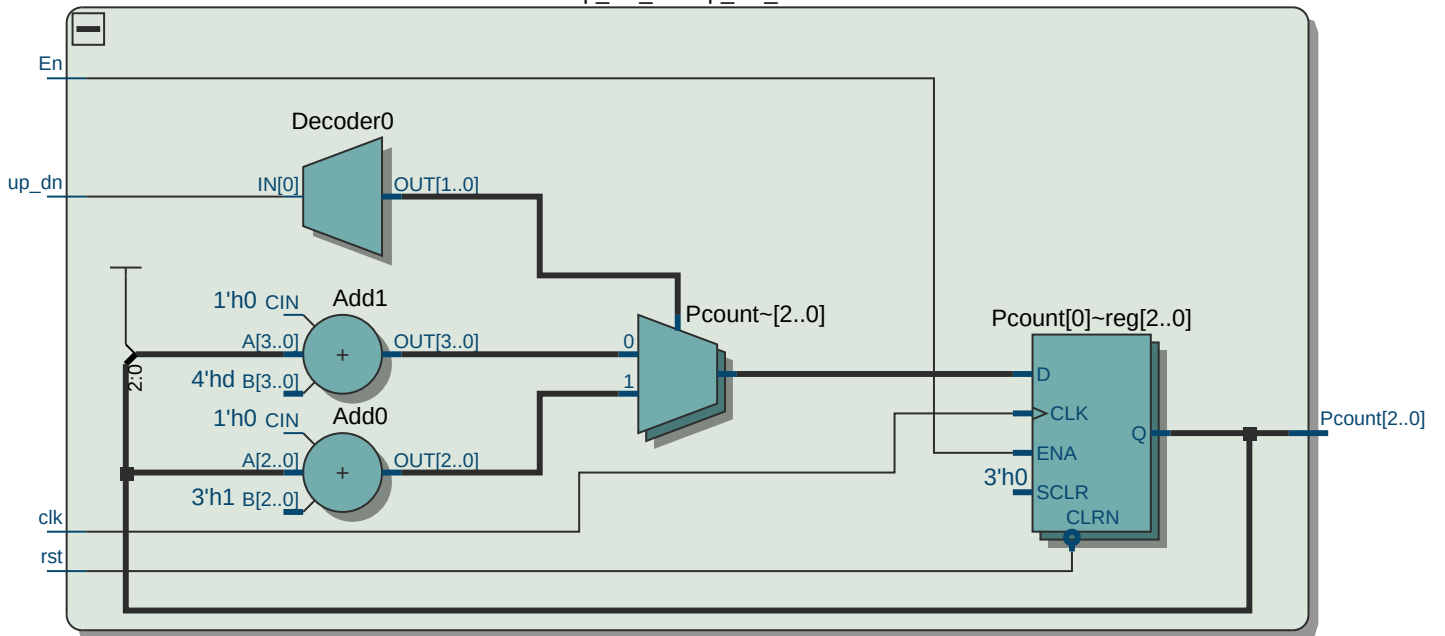
Wtime_ROM:Wtime_ROM

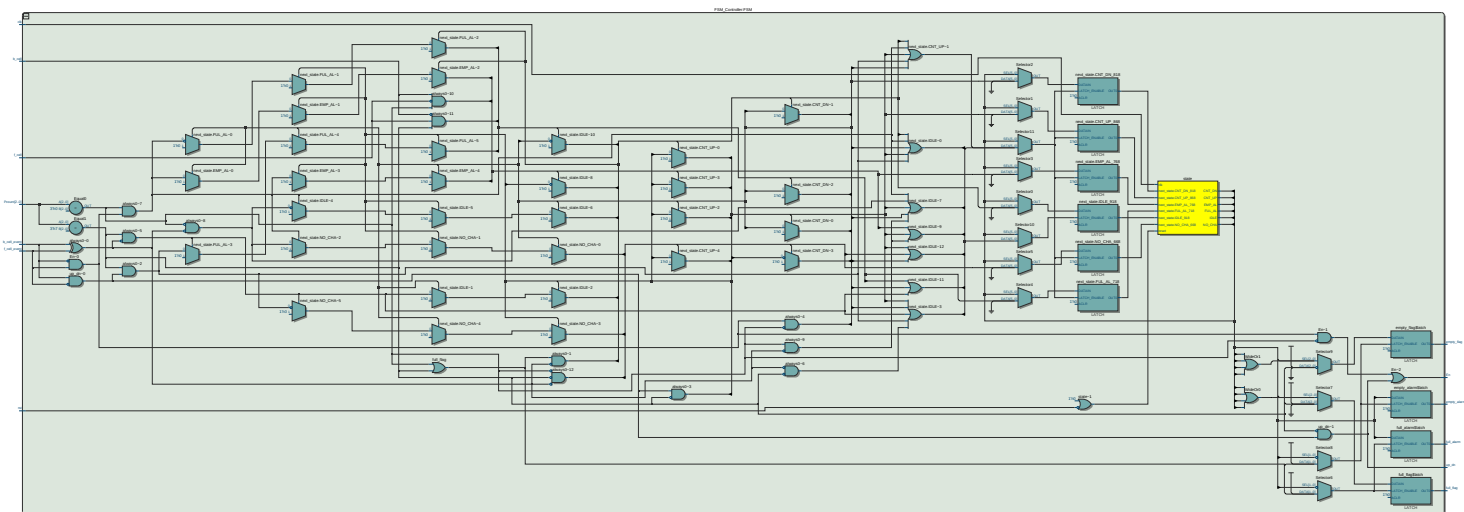


Wtime_Decoder:Wtime_Decoder



Up_Dn_cnt:Up_Dn_cnt





seven_seg_D:Pcount_Decoder

